

Oddo–Tomson · SPE 21710

Mineral-scale saturation explorer

Saturation index per scale family (calcite, sulfates, halite) swept over temperature and pressure, bottomhole → wellhead trending, and a formation-water × seawater mixing sweep for waterflood compatibility — the Oddo–Tomson conditional-solubility framework with documented, overridable coefficients.

7

scale families

WHAT YOU GET

- SI = $\log_{10}(\text{IAP} / \text{Ksp_cond})$ per family, supersaturation flagged at SI > 0
- T/P sweep and bottomhole → wellhead SI trending along the production string
- Waterflood mixing sweep (barium-rich formation water × sulfate-rich seawater)

Ask Deckhand: “Compute produced-water mineral-scale saturation indices (Oddo–Tomson) over my T/P path and check formation-water vs seawater mixing for waterflood compatibility.” — send to @the_deckhand_bot to run it live.